

Passmore Reading Brief: Is Using Venezuelan Oil Costly?

15 May 2026

Blurb: Venezuela's oil is not useless; it is just expensive to turn into something useful.

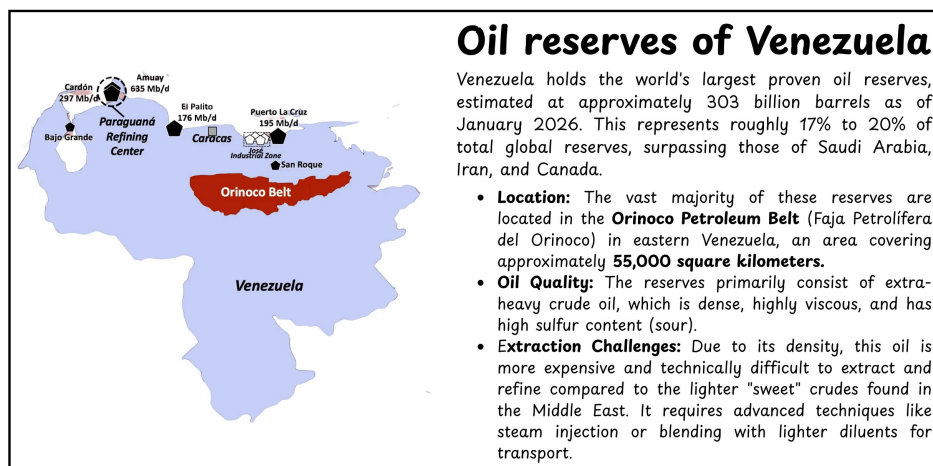
My 4 AM Reading Motivation: Trump stole it. What can he do with it?

Assertion: Venezuelan oil is abundant, but much of it is too thick, too sulfurous, and too metal-rich. Extra steps are required to make it valuable—dilution to move it, upgrading to make it refinery-ready, and blending or refining to meet final product specs. These extra steps add substantial costs, limiting the translation of Venezuela's huge reserves into the world's oil supply in proportion to the size of its reserves.

Background:

1. What Venezuelan oil is

Venezuela's oil problem is not scarcity, but quality and handling. The U.S. Energy Information Administration (EIA) says Venezuela has 303 billion barrels of proven reserves, accounting for approximately 17% of global reserves, yet it produced only 0.8% of global crude oil. Most of those reserves are extra-heavy crude from the Orinoco Belt. The U.S. Geological Survey (USGS) defines heavy oil as dense, viscous oil with an API gravity so low (the weight of oil relative to water) that it sinks in water. Also, asphaltenes in Venezuelan heavy oil contain much of the sulfur and perhaps 90% of the metals in the oil. Asphaltenes are complex hydrocarbons that can foul oil processing unless removed.



2. Why it must be blended and upgraded

Crude oil quality, in EIA's terminology, is mainly about sulfur content and API gravity, because those two properties drive processing complexity and product characteristics. In practice, Venezuelan extra-heavy crude usually is diluted with lighter hydrocarbons—natural gas liquids, light crude oil with high API gravity, or highly flammable, volatile naphtha—to move through pipelines. Diluted crude often requires chemical upgrading before further refining for final use. USGS says extra-heavy oil commonly requires diluents for transport and upgrading to reduce density and remove contaminants. Venezuelan oil producers use roughly one barrel of diluent for every three or four barrels of extra-heavy oil produced. Tariffs, trade wars, and kinetic wars create diluent shortages that can hold down Venezuela's potential oil output.



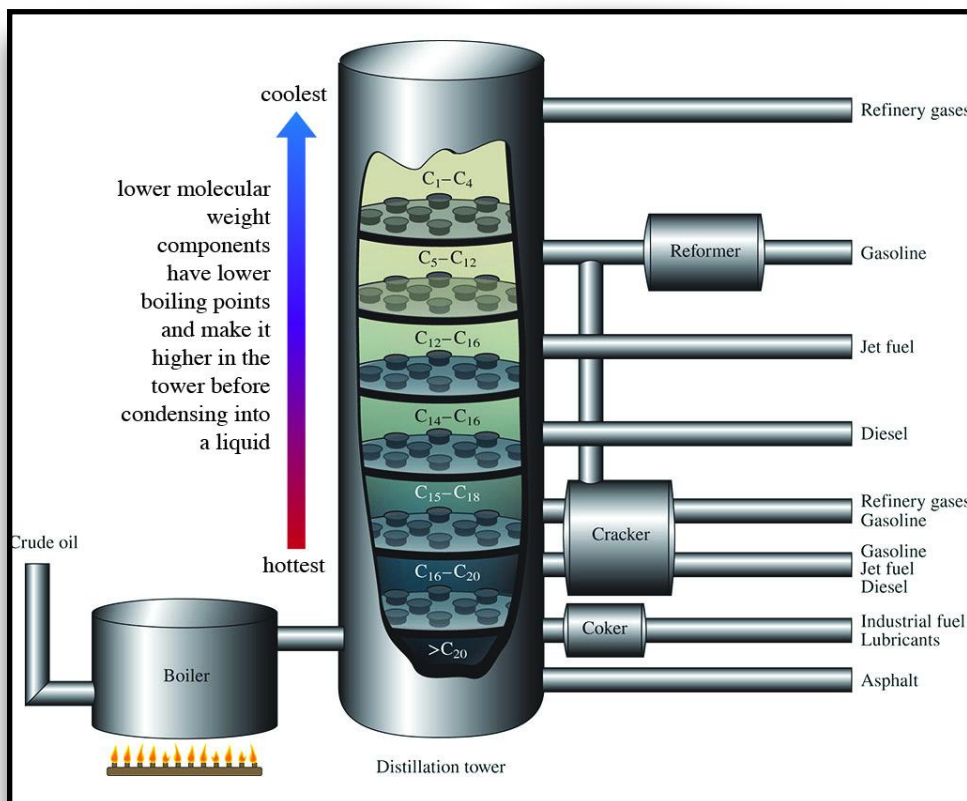
3. Who will refine it?

Once crude reaches a refinery, the challenge is to distill blended and upgraded Venezuelan oil into various petroleum products. At the same time, treatment of crude with catalysts during distillation removes sulfur and other contaminants. Venezuela has six major oil refineries. Financial sanctions, political instability, and maintenance neglect have reduced refining to about one-third of total Venezuelan capacity. Even though Chevron, an American multinational energy corporation, has maintained a presence in Venezuela, U.S. refiners with specialized infrastructure are likely to refine future Venezuelan crude exports. We are the victors, right?

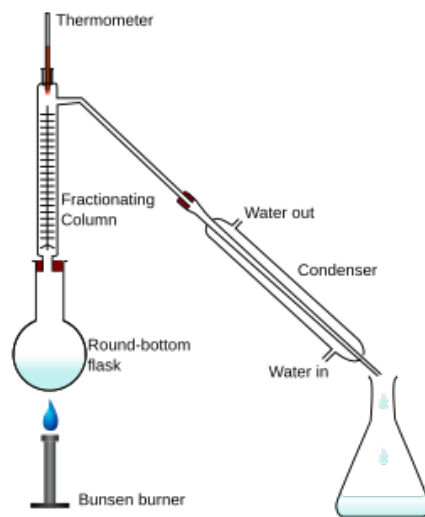


4. What refineries do to it

A refinery is a forest of distillation towers. Distillation uses heating and cooling to separate crude oil, a *mixture*, into separate components, or *fractions*. Heat evaporates oil. Fractions in the evaporated mixture differ by the temperatures and rates at which they condense from gases to liquids, according to gradients in their molecular weights. A fractionation tower distills a variety of petroleum products from crude oil.



If you are as old as I am, you might remember your chemistry teacher guiding you through a fractionation demonstration, with a stern warning not to shatter the glass of the fractionation tower as water heated to steam by a Bunsen burner met ice cold water showing how water moves through liquid and gas phases and back again. It would not surprise me if schools simulated this process on a computer instead of trusting adolescents with immature brains with hundreds of dollars' worth of glassware.



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